AFTER: Digital Twins for Cyberattack and Recovery Monitoring in Industrial Robots

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Abstract

Cyberattack detection and alarm clearance impose different demands on a robotic digital twin. A predictor can follow unauthorized motion, while an inaccurate reference can prevent clearance after recovery. This study separates physical return from reference proximity and examines three sources of error: model mismatch, measurement noise and continuing motion. In 216 synthetic episodes, thirteen monitoring rules receive common observations and approved-command context. Adding an empirical uncertainty allowance reduces benign alarm observations from 621 to 309 relative to nominal fixed return, but permits timely clearance in only 36 of 108 detected events with physical return. Removing the allowance at the same entry threshold yields 140 benign alarm observations and 108 timely clearances, with no observed premature clearing. A normal-only search over 63 reference models reduces the allowance by 51.27% and 38.56% on the harder plants without satisfying the overall feasibility requirement. A conditional Gaussian calculation explains how a noisy calibration maximum can exhaust the recovery tolerance even with a perfect reference. A subsequent covariance-aware window control meets a predeclared normal-feasibility criterion on 20 of 36 synthetic residual traces; correlation and changing motion limit the benefit of averaging. A separate common-input IMU study records improved CASPER-style training adequacy, with an ordinary autoregressive control retaining the lowest normal prediction error. These results support an evaluation discipline for recovery monitoring: distinguish prediction from physical return, account for uncertainty explicitly, and establish normal-operation feasibility before interpreting reduced alarm burden as a security improvement.

Keywords: digital twins; industrial robots; cyberattack detection; alarm clearance; system identification; model mismatch.

1 Introduction

A robot can continue to move away from its intended trajectory after a corrupted command has been removed. An alarm that clears at command restoration can therefore disappear before the physical effect ends. The reverse problem also occurs: an accurate account of the robot's recovery may remain inconsistent with an inaccurate nominal model, leaving an alarm active after normal motion resumes. Both errors matter when a digital twin supplies evidence for cyberattack monitoring.

Recent robotic anomaly detectors use motion measurements to identify departures from expected behavior [1, 2]. Hybrid twins combine models of robot mechanics, drives and control with learned components [3]. These approaches motivate a closer examination of the relationship between prediction quality and alarm recovery. A small one-step prediction error need not indicate a return to authorized motion, because the predictor may already be following the deviation. Conversely, a large free-reference error can reflect ordinary model mismatch rather than a continuing attack.

We study a specific question: *under model mismatch, when does discrepancy from a command-driven reference provide usable evidence for clearing a robotic attack alarm?* The monitor receives noisy joint positions and velocities together with a protected record of approved commands. An attacker can corrupt the downstream command path. This information model permits a reference to predict authorized motion while preserving a distinction between approved and executed commands.

The investigation has three connected stages. A prospectively specified attack evaluation compares thirteen detection and clearance rules under common observations. A subsequent normal-only study tests whether better clean-data identification restores clearance feasibility. A Gaussian analysis and a separately specified residual experiment then examine noise allocation, temporal correlation and the cost of bounding current rather than average motion. Later stages diagnose the initial failure without revising its attack results. A separate external-IMU prediction study, reported in Appendix B, checks the training adequacy of recent architectural comparators without giving them access to the simulated command context.

The contribution is an experimentally supported account of a failure mode in uncertainty-based alarm clearance. It combines a physical-return target, matched component comparisons, normal-reference development and explicit noise calculations. The window experiment further identifies when averaging restores a usable return margin and when its motion penalty prevents that benefit. The bounds use established interval and probability arguments. AFTER names this investigation; it does not denote a newly validated detector or a claimed improvement over complete recent systems.

2 Related work

2.1 Robotic anomaly detection and digital twins

CASPER detects anomalous industrial-arm motion using external inertial sensing and contextual information [1]. Its edge-computing successor develops real-time robotic anomaly detection [2]. Their sensing modality differs from the joint-state and approved-command simulation here. Appendix B evaluates independently written architectural implementations on the same nominal IMU observations, with causal preprocessing and ordinary controls. This separate study supports a normal prediction comparison; it does not reproduce either paper's attack benchmark or justify ranking IMU-only predictors against command-aware recovery monitors.

Ayankoso et al. describe a hybrid twin combining robot kinematic and dynamic models, drive and control information, and learned uncertainty accommodation for collaborative-robot condition monitoring [3]. This establishes the relevance of model fidelity and uncertainty to robotic twins. It does not establish that an uncertainty allowance improves attack-alarm clearance. More broadly, Classens et al. develop recursive identification of structured mechanical systems through instrumental variables and block-coordinate descent [4]. Our fixed ARX/FIR search is an ordinary identification exercise rather than an alternative claim to that contribution.

2.2 Uncertainty, runtime monitoring and recovery

FAIL-Detect learns or constructs scalar signals and calibrates uncertainty-aware failure monitors from successful robot rollouts [5]. Its full system concerns imitation-learning policies and includes signal-learning and calibration choices absent from our experiment. We independently implement a restricted constant episode-band component on our shared discrepancy signal. We therefore compare a thresholding principle under common inputs, not the full published method.

Interval-observer fault detection explicitly considers the effect of uncertain models and observer design [6]. Adding a bound to a measured residual, or requiring a dwell period before clearing an alarm, is not by itself a new digital-twin mechanism. The useful question here is whether the resulting decision remains feasible at the tolerance required by the recovery definition.

Kaisar et al. describe a robotic-vehicle architecture combining statistical detection, a kinematic twin, supervisory handover and consistency-gated recovery [7]. The available abstract establishes direct overlap with generic twin-assisted recovery; its full algorithm and implementation were unavailable for an equation-level comparison. Time-uniform inference is also established: Howard et al. develop confidence sequences under explicit probabilistic assumptions [8]. The window control below uses elementary Gaussian tails and error spending, with no claim to reproduce those sharper constructions or introduce a new confidence-sequence method.

The RESCUE preprint evaluates the value of runtime alarms through counterfactual robot-policy interventions [9]. It shows why detection accuracy and intervention usefulness must be distinguished. Our evaluator instead compares corrupted and uncorrupted executions of the same plant to measure the disappearance of a motion deviation. The overlap is substantive: counterfactual evaluation is existing methodology. Our narrower emphasis is the interaction among reference error, sensor noise and a particular alarm-clearance rule. The experiments support conclusions about this rule and these simulated conditions, rather than a ranking against complete recent robotic-security systems.

3 Recovery target and clearance feasibility

3.1 What the monitor observes

Let $x_t = (q_1, q_2, \dot{q}_1, \dot{q}_2)$ denote the actual state and x_t^0 the state of the same plant under the complete approved command history without corruption. The monitor receives $y_t = x_t + n_t$, approved commands u_t, and their availability indicator. A reference r_t predicts the authorized trajectory. Positive scales s_j are estimated from clean training observations only. Define

$$\|v\|_s = \max_j |v_j|/s_j, \qquad d_t = \|x_t - x_t^0\|_s, \qquad c_t = \|y_t - r_t\|_s. \tag{1}$$

The evaluator computes d_t; the monitor computes c_t. In the noise-free case, $x_t = r_t \neq x_t^0$ can give $c_t = 0$ with $d_t > 0$. Similarly, $x_t = x_t^0 \neq r_t$ gives $d_t = 0$ with $c_t > 0$. A reference-discrepancy threshold consequently requires an assumption linking r_t to x_t^0 before it can represent physical return.

3.2 Conditional error bounds

Proposition 1 *If $\|x_t^0 - r_t\|_s \leq b_t$ and $\|n_t\|_s \leq \eta_t$, then*

$$\max(0, c_t - b_t - \eta_t) \leq d_t \leq c_t + b_t + \eta_t. \tag{2}$$

Proof. Write $x_t - x_t^0 = (y_t - r_t) - n_t - (x_t^0 - r_t)$ and apply the triangle and reverse triangle inequalities. Aligned scalar errors attain the extremes, so scalar allowances alone do not generally permit tighter bounds. *Q.E.D.*

A sufficient return condition at tolerance δ is $c_t + b_t + \eta_t \leq \delta$. This gives a simple feasibility test before any alarm experiment.

Proposition 2 *Suppose reference and measurement errors lie in coordinate boxes with half-widths $b_{t,j}$ and $\eta_{t,j}$. Return within δ holds for every pair in those boxes exactly when*

$$\max_j \frac{|y_{t,j} - r_{t,j}| + b_{t,j} + \eta_{t,j}}{s_j} \le \delta. \tag{3}$$

If $b_{t,j} + \eta_{t,j} > \delta s_j$ in any coordinate, no observation can satisfy this condition.

Proof. In each coordinate, the two admissible errors can align against the measured displacement. Their summed magnitude attains the triangle-inequality bound. The observed absolute displacement is nonnegative, yielding the infeasibility statement. *Q.E.D.*

These statements are conditional geometric facts. The simulated sensors have unbounded Gaussian noise, and the experiment uses an empirical discrepancy allowance rather than a verified deterministic enclosure. Specifically, the tested rule requires $c_t + b \le \delta$, where b is the largest observed normal discrepancy. It is a heuristic motivated by the upper-bound form, not an implementation of a proven safety certificate. Because b contains both model error and noise, interpreting it as pure reference uncertainty would be incorrect. A dwell period cannot restore feasibility when $b > \delta$.

4 Simulation and threat model

4.1 Plant, controller and sensing

All observations in the attack and recovery simulations are synthetic. The separate IMU prediction study uses a received public nominal recording and is described independently in Appendix B. The simulated plant is a horizontal two-joint point-mass arm governed by

$$M(q)\ddot{q} + C(q, \dot{q}) + B\dot{q} = \tau. \tag{4}$$

Writing $a = m_2 l_1 l_2$ and $d = m_2 l_2^2$,

$$M(q) = \begin{bmatrix} (m_1 + m_2)l_1^2 + d + 2a\cos q_2 & d + a\cos q_2 \\ d + a\cos q_2 & d \end{bmatrix}, \tag{5}$$

$$C(q, \dot{q}) = a \sin q_2 \begin{bmatrix} -2\dot{q}_1\dot{q}_2 - \dot{q}_2^2 \\ \dot{q}_1^2 \end{bmatrix}. \tag{6}$$

A PD controller requests componentwise saturated torque

$$\tau^* = \mathrm{clip}\{K_p(q^c - q) + K_d(\dot{q}^c - \dot{q}), -20, 20\}, \tag{7}$$

with $K_p = \mathrm{diag}(18, 12)$ and $K_d = \mathrm{diag}(4, 3)$. Where actuator lag is present, $\dot{\tau} = (\tau^* - \tau)/T_a$. Nominal masses are $(1, 0.8)$ kg, lengths $(0.5, 0.4)$ m, and damping coefficients $(0.08, 0.06)$. The nominal physics reference retains these parameters; Table 1 specifies the actual plants.

Position and velocity noise are independent zero-mean Gaussians with standard deviations 0.0005 rad and 0.002 rad/s, respectively, before the plant multiplier. Episodes contain 1,200 observations at 50 Hz. Fourth-order Runge–Kutta integration uses eight substeps per observation. A short lagged trajectory was checked against a separate SciPy integration. This verifies numerical consistency; the model excludes gravity, contact, flexibility and thermal effects and has not been validated against physical robot measurements.

Table 1: Simulated plant configurations. Mass and damping entries multiply the nominal values; noise multiplies the stated sensor standard deviations.

Plant	Mass factors	Damping factors	Lengths (m)	Lag (s)	Noise
Q1	(1.10,0.86)	(0.80,1.20)	(0.48,0.42)	0	1.0
Q2	(0.88,1.15)	(1.30,0.85)	(0.53,0.37)	0.03	1.0
Q3	(1.16,1.06)	(0.90,1.40)	(0.46,0.43)	0.06	1.5

4.2 Adversary and information boundaries

The attacker changes executed commands downstream of the protected approved-command record. It can impose an offset, slow progression or hold a command. Joint measurements remain truthful apart from the declared sensor noise. Legitimate task changes update both command paths. The monitor receives no intervention labels, true state, actual plant parameters or paired counterfactual trajectory in its calculations.

The attack-study implementation passes a complete episode dictionary to the monitoring routine. Source inspection and three restricted-input replays support the stated input use, but this is not an enforced isolation boundary. The later normal-only study uses a restricted observation type. Neither study addresses joint compromise of sensing and approved commands, poisoned commissioning, or attribution of malicious intent from motion alone.

Missing approved commands invalidate the complete history required by a free reference. Command-dependent methods therefore remain unknown for the rest of that episode, even after availability resumes. Unknown intervals remain in coverage and missed-event accounting; they are not silently removed from the evaluation population. Persistence does not require commands.

5 Study I: attack detection and recovery

5.1 Clean commissioning and model state

Each plant provides three known-clean commissioning episodes. Their eligible observations define the channel means and population standard deviations. Let $z_t = (y_t - \mu)/s$ denote standardized state. The ARX reference predicts

$$\widehat{z}_t = a_0 + \sum_{k=1}^{p} A_k \widetilde{z}_{t-k} + \sum_{k=1}^{p} D_k u_{t-k}, \tag{8}$$

where the four-component command contains approved position and velocity. For the free reference, $\widetilde{z}$ is its own predicted state after initialization. The one-step control instead uses measured state history. Thus the reference's maintained state is its lagged trajectory; its command-driven recurrence makes prospective predictions, while synchronization is restricted to the initial observed prefix. The following observer provides a separate measurement-corrected comparison.

Study I fixes $p = 4$ and fits a ridge of 10^{-7} to mean normal equations, without penalizing the intercept. The first 100 observations are excluded from fitting targets. Homogeneous recursion must have spectral radius below one. The free reference starts from four observed states and receives no subsequent measurement correction. Fitted parameters remain fixed during evaluation.

5.2 Monitoring rules and calibration

All scores are maximum absolute standardized discrepancies. The roster includes persistence, one-step ARX, a nominal following observer, nominal free references, and learned free references. The following observer predicts before correcting its state with gains $(0.65, 0.65, 0.35, 0.35)$. Point rules alarm when their score exceeds h. Hysteresis enters at the same strict exceedance and clears after ten valid observations at or below $h/2$.

Fixed-return rules instead clear after ten observations with the relevant reference discrepancy at most $\delta = 0.1$. The uncertainty-return candidate requires $c_t + b \leq 0.1$. An entry-threshold exceedance takes priority and resets the clearance counter. Two matched ablations keep the candidate entry threshold while removing b or reducing the dwell from ten samples to one.

Each plant has nineteen normal calibration episodes. Each main rule calibrates its complete alarm state machine to at most 209 alarmed observations among 20,900 eligible observations. The threshold search selects the boundary score satisfying this finite budget. The candidate allowance is the largest learned-reference discrepancy in the same calibration episodes. Calibration uses no attack labels and does not guarantee a future 1% false-alarm rate.

The additional constant episode band takes the maximum of the nineteen whole-episode discrepancy maxima. Its rank is $\lceil (19+1)(1-0.05) \rceil = 19$. This component has a different calibration objective from the main point-alarm budget. The fixed calibration grid and shifted task families do not establish the exchangeability needed for a conformal coverage guarantee. Matched-entry ablations also have their own realized calibration burdens; their role is to isolate components at an identical entry threshold.

5.3 Frozen evaluation and metrics

The fresh matrix crosses three new command families, three plants, seeds 17, 42 and 2026, and eight conditions, giving 216 episodes and 2,808 method/episode outcomes. The conditions are unchanged motion, authorized offset, authorized slowdown, brief unauthorized offset, persistent offset, held command, unauthorized slowdown, and offset with missing command context. The first 100 observations are processed but excluded from metrics. Corruption begins at sample 320 and ends before sample 620, except the brief offset, which lasts six samples. Offset magnitude is 0.14 rad on joint 1, slowdown rate is 0.6, and missing context spans $[320, 380)$. States do not reset at intervention boundaries.

For evaluation only, a paired simulation uses the same plant and initial state with approved commands throughout. Physical return is the first post-restoration run of ten true-state observations satisfying $d_t \leq 0.1$, searched over five seconds. Its timestamp is the start of that qualifying window, assigned retrospectively by the evaluator. This tolerance defines a motion-effect diagnostic, not task success, collision avoidance or product quality. Nominal-reference return is retained as a secondary diagnostic.

We report unadjusted precision, recall and F1 against the command-corruption interval, average precision (AP), alarm observations and episodes on wholly benign trajectories, coverage, event overlap, fresh alarm onset, misses, premature clearing and timely clearance after physical return. An alarm already active at intervention onset counts as overlap but not fresh detection. Persistent post-intervention alarms can be false positives under interval labels while still reflecting unrecovered motion; the recovery measures expose this difference. Conditional clearance counts include only detected events with measured physical return, while all misses remain visible.

Premature clearing means an active-to-inactive transition after first detection and before the physical-return timestamp. Timely clearance means a known inactive alarm observation within

five seconds of that timestamp, including alarms already inactive when physical return begins. Consequently, the timely-clearance count must be interpreted alongside premature clearing; it does not by itself show that an alarm remained active until recovery.

The predeclared primary criterion combines four requirements: at least 25% fewer benign alarm observations than nominal fixed return; no additional missed or prematurely cleared events; at least 95% timely clearances among detected-returned events; and benign burden no more than 10% above learned-reference hysteresis. Models and thresholds were sealed before evaluation generation, and all score artifacts were sealed before joint outcome computation.

5.4 Results

The primary criterion failed. The candidate reduced benign alarm observations from 621 to 309, or 50.24%, but achieved timely clearance in only 36 of 108 detected-returned events. It overlapped 108 of 135 interventions, with 104 fresh onsets. Table 2 retains every method; Figure 1 contrasts its apparent alarm-burden benefit with the clearing failure.

Table 2: Study I: complete pooled comparison. Benign counts use 89,100 observations. Overlap and fresh onset each use 135 interventions. Early counts events with premature clearing. Clear/return retains the detected-returned denominator. CP and matched-entry rows have distinct calibration interpretations.

Method	Benign	F1	Overlap/fresh	Early	Clear/return
Persistence	878	0.057	127/127	127	127/127
ARX one-step	792	0.803	108/108	108	108/108
Following point	794	0.808	108/108	108	108/108
Following hysteresis	993	0.785	108/108	108	108/108
Nominal hysteresis	789	0.713	108/108	45	108/108
Nominal fixed return	621	0.657	108/108	0	108/108
Learned point	819	0.700	108/108	18	108/108
Learned hysteresis	792	0.683	108/108	0	108/108
Learned fixed return	827	0.690	108/108	0	108/108
Uncertainty return	309	0.446	108/104	0	36/108
Episode CP band	0	0.744	108/108	75	108/108
Matched no allowance	140	0.709	108/108	0	108/108
Matched no dwell	20	0.453	108/108	0	36/108

The learned-reference spectral radii are 0.95595, 0.96690 and 0.97148 for Q1–Q3. Their empirical allowances are 0.07088, 0.19149 and 0.20928. Thus Q2 and Q3 have $b > \delta$, making clearance mathematically impossible for the tested rule. Their zero benign alarm counts do not demonstrate useful recovery. All 36 timely candidate clearances come from Q1.

Removing the allowance at the same entry threshold reduces benign alarms further, to 140, and restores timely clearance in all 108 detected-returned events. Fresh onsets increase from 104 to 108, with no observed premature clears. Removing dwell alone reduces benign alarms to 20 but leaves the same 72 clearing failures. The matched comparison identifies the allowance as the obstruction; a longer or shorter waiting period cannot remove it.

The following-observer point rule achieves F1 0.808 but clears prematurely in all 108 detected events. The episode band has no benign alarm observations and F1 0.744, yet clears prematurely in 75 events. Learned-reference hysteresis has 792 benign alarms, no premature clears and 108 timely clearances. These outcomes show why interval detection scores and recovery measurements must be read together.

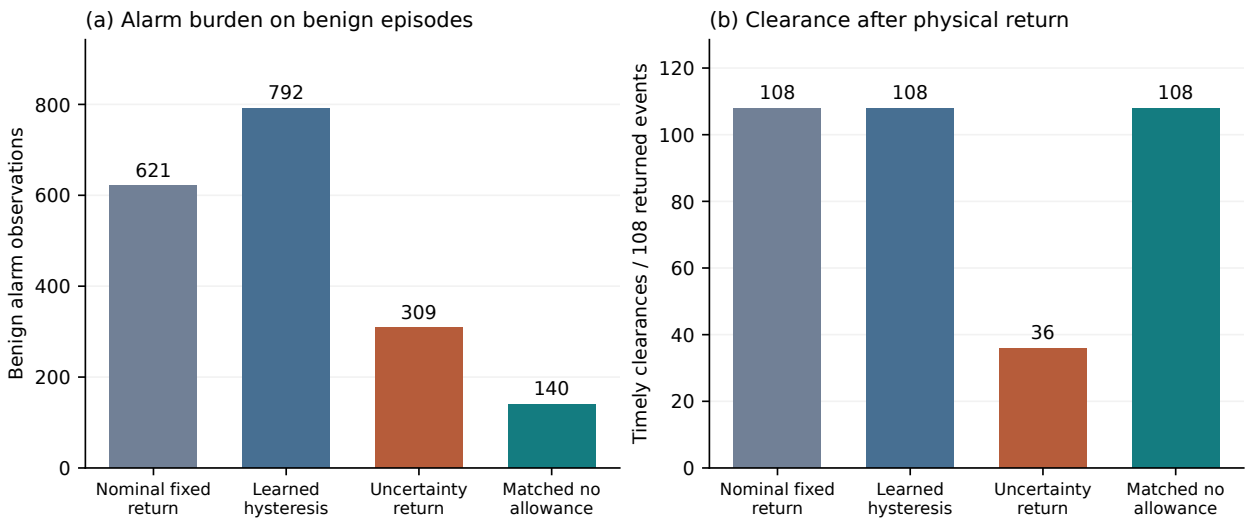


Figure 1: Saved Study I outcomes. All four displayed rules have zero observed premature clears, but the uncertainty-return rule fails timely clearance on 72 detected-returned events. Fewer benign alarm observations alone conceal this failure.

All 27 missing-context interventions are missed by command-dependent methods under the unknown-history rule. Nominal-reference and paired-plant return times differ in 21 of 135 intervention episodes. The empirical allowance encloses the complete eligible counterfactual-reference discrepancy in 189 of 216 episodes, which does not establish a distribution-shift guarantee. Further point metrics, unknown coverage and alarm-episode rates appear in Appendix C.

6 Study II: can better commissioning restore feasibility?

6.1 Finite normal-only development procedure

Study II follows Study I and is an outcome-informed diagnosis. It uses the same three clean commissioning episodes per plant but introduces a finite model bank before generating new normal checks. ARX orders $\{4, 8, 16, 32\}$ and FIR lengths $\{16, 32, 64\}$ each use ridge values $\{10^{-7}, 10^{-5}, 10^{-3}\}$. There are 21 fits per plant and 63 in total. All are eligible; no failed candidate is omitted. FIR uses the command-history terms and intercept in Equation 8, without state-history terms.

The nineteen previously used normal calibration episodes are explicitly reused for development: the first nine select a model, the next five estimate its allowance, and the last five provide archived checks. Selection minimizes the worst episode's 99th-percentile discrepancy, breaking ties by mean standardized RMSE, parameter count and model identifier. The allowance is the maximum over the five assigned episodes, or 5,500 observations. The archived checks are not relabeled independent holdouts.

Every reference receives the same 100-observation initialization prefix. A free ARX model uses the last p states of that prefix, then feeds back its predictions. This differs from Study I's four-state initialization and nineteen-episode allowance. We therefore compare selected models to a matched ARX(4) within Study II. Its fitted coefficients and training transformations equal the earlier fit exactly; differences between studies' allowances cannot be attributed solely to model order.

New checks cross three previously ungenerated task families, three plants, three seeds and three normal conditions: unchanged motion, authorized offset and authorized slowdown. This gives 81

new normal episodes. Including fifteen archived checks and six recorded roles gives 576 outcomes. The selected overall model equals the selected ARX on each plant; these duplicate roles are not separate evidence.

6.2 Feasibility criterion

For each reference, we measure the fraction of eligible normal observations satisfying $c_t + b \leq 0.1$. Four diagnostic probes per episode, starting at samples 100, 320, 620 and 900, force an active latch and seek ten qualifying observations within five seconds. These probes ignore alarm-entry priority: they test the opportunity to clear during normal motion, not attack detection or actual post-attack recovery.

The predeclared admission rule requires $b < 0.1$, at least 95% feasible normal observations, and at least 95% successful probes on every plant. A model can clear every probe yet fail the point-fraction condition. The criterion is retained unchanged after observing the results.

6.3 Results

The selected ARX orders are 8, 16 and 32 for Q1, Q2 and Q3, all with ridge 10^{-7}. Mean standardized RMSE falls from 0.009477 to 0.009337 on Q1, from 0.010203 to 0.009716 on Q2, and from 0.015865 to 0.014388 on Q3. Table 3 and Figure 2 show that these improvements do not establish clearance feasibility across all plants.

Table 3: Study II: selected ARX versus matched ARX(4). b_4 and $b_\star$ are their respective allowances. Feasibility uses 29,700 new normal observations per plant. Each plant has 108 diagnostic probe starts.

Plant	Order	b_4	$b_\star$	Reduction (%)	Feasible (%)	Clear/probes
Q1	8	0.0553	0.0553	0.06	99.82	108/108
Q2	16	0.1469	0.0716	51.27	93.32	108/108
Q3	32	0.1913	0.1175	38.56	0.00	0/108

On Q2, the allowance drops 51.27%, from 0.146910 to 0.071592. All 108 probes clear, yet only 93.32% of normal observations satisfy the inequality. Q3 improves by 38.56%, from 0.191250 to 0.117495, but retains an allowance above the tolerance and clears none of its probes. Only Q1 passes all requirements. FIR(64) is selected within the FIR family on all plants but performs worse in this search. The full role comparison, including one-step ARX and command-as-state controls, is in Appendix D. No attack evaluation of the selected replacements was conducted.

7 Noise allocation explains the remaining obstruction

The normal-study outcome motivates a separate post-study calculation. Consider a hypothetical perfect free reference $r_t = x_t^0$ under normal motion, with the simulator's independent Gaussian measurement noise and fixed training scales. Let $\nu_j = \sigma_j / s_j$. The distribution of the scalar noise discrepancy is

$$F(x) = \Pr\{\|n_t\|_s \leq x\} = \prod_{j=1}^{4} \operatorname{erf}\left(\frac{x}{\sqrt{2}\nu_j}\right), \qquad x \geq 0. \tag{9}$$

For an empirical allowance B equal to the maximum of N independent calibration discrepancies,

$$\Pr(B \leq b) = F(b)^N. \tag{10}$$

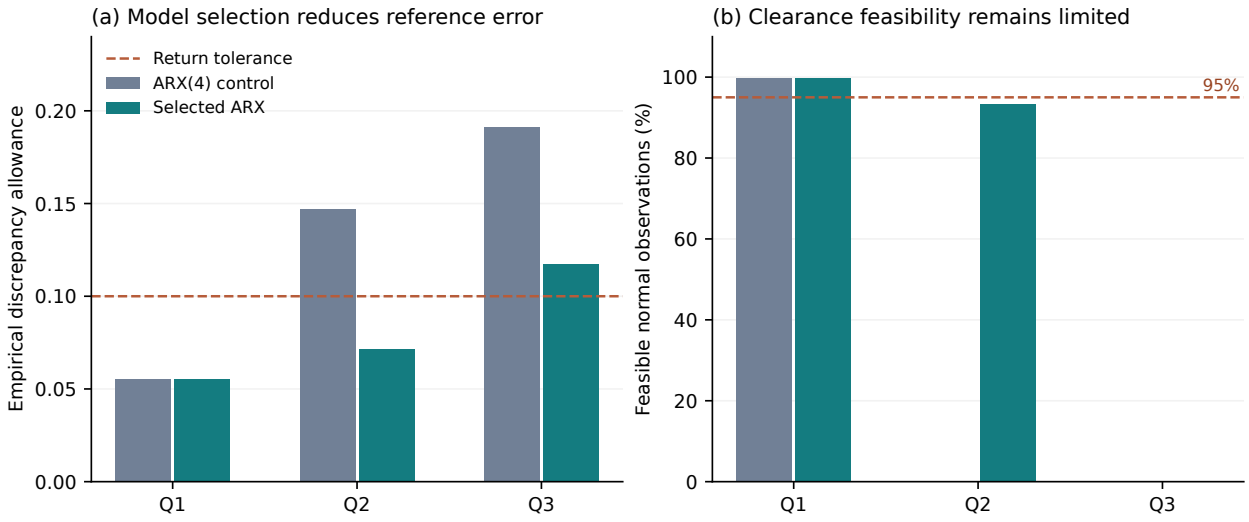


Figure 2: Normal-only model development. Lower discrepancy allowances make Q2's inequality reachable, but its feasible fraction remains below 95%. Q3's selected allowance still exceeds the return tolerance.

Conditioned on B, a new normal observation satisfies the rule with probability $F(\delta - B)$, taking $F(x) = 0$ for $x < 0$. A population feasibility of at least γ therefore requires

$$B \le \delta - F^{-1}(\gamma), \qquad \Pr\{F(\delta - B) \ge \gamma\} = F\big(\delta - F^{-1}(\gamma)\big)^N. \tag{11}$$

This elementary calculation describes allocation of a fixed tolerance between two noisy quantities. Increasing N makes the calibration maximum stochastically larger, even though a larger calibration sample might otherwise seem reassuring. Better reference fitting cannot remove this noise-only contribution.

Table 4: Conditional perfect-reference diagnostic with $N = 5{,}500$, $\delta = 0.1$ and $\gamma = 0.95$. The final column is the probability that a noise-only calibration maximum leaves enough margin for 95% population feasibility.

Plant	$F^{-1}(0.95)$	$\delta - F^{-1}(0.95)$	Margin probability
Q1	0.028251	0.071749	0.999828
Q2	0.027914	0.072086	0.999897
Q3	0.041737	0.058263	2.19×10^{-9}

For Q3, the single-observation noise quantile is 0.041737, leaving an allowable calibration maximum of only 0.058263. Equation 11 gives approximately 2.19×10^{-9}. The corresponding probabilities for Q1 and Q2 exceed 0.9998. Two numerical implementations agree on these values.

The calculation conditions on the declared independent Gaussian noise and the fitted training scales. It is a population statement about the clearance inequality, not the probability of passing a finite empirical check, a dwell-time calculation, or an estimate across real robots. It was added after Study II and changed no selected model, tolerance or criterion. It explains why the maximum-plus-discrepancy construction can remain unusable even when reference error is removed; it does not establish that all uncertainty-aware recovery decisions fail.

8 Study III: covariance-aware return bounds

8.1 A standard control for current deviation

The noise-only obstruction motivates a control that averages noise while retaining a bound on current motion. This section states the assumptions explicitly, rather than interpreting a noisy empirical maximum as a model-error enclosure. For a dimensionless channel j, write

$$e_{t,j} = v_{t,j} + \beta_{t,j} + \epsilon_{t,j}, \qquad |\beta_{t,j}| \le B_j, \quad |v_{t,j} - v_{t-1,j}| \le L_j, \tag{12}$$

where v is physical deviation from the nominal response, β is reference bias, and ϵ is measurement noise. Assume a stationary zero-mean Gaussian AR(1) noise process with marginal standard deviation σ_j and correlation $0 \le \rho_j < 1$. The variance of a mean over m consecutive noise observations is

$$V_{m,j} = \frac{\sigma_j^2}{m^2}\left[m + 2\sum_{k=1}^{m-1}(m-k)\rho_j^k\right]. \tag{13}$$

Independence gives $V_{m,j} = \sigma_j^2/m$. Positive serial correlation reduces the effective averaging benefit.

Let $\mathcal{M}$ contain K fixed windows, let d be the number of channels, and allocate $\alpha_t = 6\alpha/(\pi^2 t^2)$ using a global clock that does not restart at alarms. Define

$$U_{t,j}(m) = |\overline{e}_{t,j}(m)| + B_j + \sqrt{2V_{m,j}\log(2dK/\alpha_t)} + \frac{L_j(m-1)}{2}. \tag{14}$$

The last term is essential: a small window-average deviation need not imply a small deviation now.

Proposition 3 *Under the stated Gaussian covariance, bias and increment assumptions, with probability at least* $1-\alpha$, $|v_{t,j}| \le U_{t,j}(m)$ *simultaneously for every channel, complete registered window and prospective time within a trace. Consequently, the decision*

$$\min_{m\in\mathcal{M}} U_{t,j}(m) \le \delta_j \qquad (j = 1, \dots, d) \tag{15}$$

certifies coordinatewise return on that simultaneous event.

Proof. The Gaussian mean has variance $V_{m,j}$, so its two-sided tail at the square-root term in Equation 14 is at most $\alpha_t/(dK)$. Sum over channels, registered windows and times, using $\sum_{t=1}^{\infty}\alpha_t = \alpha$. Window overlap and cross-channel dependence do not invalidate this union bound. Finally, $|\overline{\beta}| \le B_j$ and $|v_{t,j} - \overline{v}_{t,j}(m)| \le m^{-1}\sum_{k=0}^{m-1} kL_j = L_j(m-1)/2$. The triangle inequality gives the result. Each coordinate may use a different minimizing window because all bounds hold simultaneously. *Q.E.D.*

This is an elementary comparison control, not a new time-uniform inference result. Its error allocation is conservative, and its validity depends on the supplied covariance and deterministic bounds. Unknown or invalid observations invalidate affected windows; they do not reset the clock. An absence of a certificate indicates uncertainty, not an attack alarm. The calculation does not model arbitrary colored, heavy-tailed or nonstationary noise.

8.2 Prospective residual experiment

The diagnostic fixes $d = 4$, $\mathcal{M} = \{1, 41, 201\}$, $\alpha = 0.05$, $\delta_j = 0.1$ and $B_j = 0.01$. Four (σ, ρ) settings are $(0.03, 0)$, $(0.06, 0)$, $(0.03, 0.8)$ and $(0.06, 0.95)$, with channel standard-deviation multipliers

$(1, 0.8, 1.1, 0.9)$. For phases $\phi_j = (0, 0.4, 0.8, 1.2)$, bias is $0.01\cos(0.003t + \phi_j)$. The physical-deviation signals are zero, $0.02\sin(0.004t + \phi_j)$, and $0.02\sin(0.04t + \phi_j)$, termed settled, slow and faster variation. They always lie within tolerance. The corresponding increment bound is $L_j = 2a|\sin(\omega/2)|$, derived from the declared amplitude and frequency.

Each trace contains 1,200 samples, with the first noise value drawn from the stationary marginal. Seeds 17,42,2026 produce three independent innovation arrays reused across settings, giving 36 paired constructed traces rather than 36 independent acquisitions. The common reporting support is samples 201–1,200, or 1,000 endpoints per trace. The error statement is marginal per trace; no simultaneous 95% guarantee across this whole matrix is claimed.

All four valid controls—the three fixed windows and their simultaneous envelope—receive identical residuals and parameter bounds. Two deliberately misspecified controls set $\rho = 0$ for the 41- and 201-sample windows. Every variant and all 216 outcomes are retained. The criterion requires at least 95% of common endpoints certified on every trace by the envelope. Parameters and conditions were recorded before generating outcomes; none was changed afterward.

These are synthetic residual signals, not outputs of the robot dynamics in Section 4. Noise, bias and increment bounds are known generator parameters. The experiment isolates their mathematical effects without establishing that those bounds are available from commissioning data. There is no attack, threshold fitting, model learning or measured post-attack recovery delay.

8.3 Results and implications

The envelope meets the criterion on 20 of 36 traces, so the overall normal-feasibility requirement fails. Table 5 reports all condition groups, including seed ranges; Figure 3 displays each group's worst seed. No instantaneous control endpoint is certified. Averaging restores normal certificates for several settings, but neither the longest window nor the assumption of independent noise is uniformly appropriate.

Table 5: Study III: simultaneous-envelope certificates on normal synthetic residuals. Each range spans the three seeds, each with 1,000 common endpoints. Passing requires at least 95% on each trace.

Noise setting	Deviation profile	Certificates (%)	Passes
Independent, low	Settled	100.0–100.0	3/3
Independent, low	Slow variation	100.0–100.0	3/3
Independent, low	Faster variation	100.0–100.0	3/3
Independent, high	Settled	100.0–100.0	3/3
Independent, high	Slow variation	100.0–100.0	3/3
Independent, high	Faster variation	17.2–25.6	0/3
Correlated, low	Settled	100.0–100.0	3/3
Correlated, low	Slow variation	85.8–100.0	2/3
Correlated, low	Faster variation	0.0–0.0	0/3
Correlated, high	Settled	0.0–0.0	0/3
Correlated, high	Slow variation	0.0–0.0	0/3
Correlated, high	Faster variation	0.0–0.0	0/3

For independent low noise with faster variation, window 41 certifies every endpoint, whereas window 201 certifies none. The longer window's drift allowance consumes its noise benefit. Under correlated low noise and slow variation, the envelope's fractions are 100%, 85.8% and 100%: their 95.27% average does not satisfy the all-traces criterion. At $\rho = 0.95$ and $\sigma = 0.06$, no valid control certifies any common endpoint, even for the settled profile. The misspecified 201-sample control

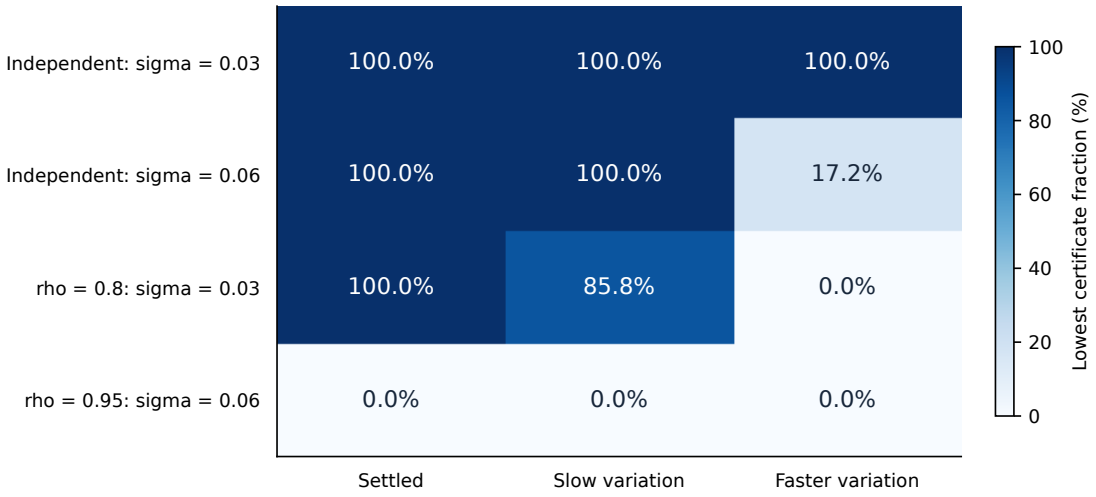


Figure 3: Lowest certificate fraction across seeds for each synthetic residual condition. All physical-deviation signals are within tolerance. A certificate establishes only the conditional bound in Equation 15; low coverage is not an attack-detection result.

would certify 91.53% on average for that settled profile, but its uncertainty claim no longer follows from the actual covariance.

No coverage violation is observed in the within-tolerance traces. This is a numerical check under a constructed normal distribution, not an estimate of rare unsafe-clearing risk or evidence near the tolerance boundary. Longer averaging reduces the noise term while increasing the current-state variation term. A useful recovery mechanism must resolve that trade-off with justified information about noise and motion, rather than selecting the most favorable observed window after evaluation.

The diagnostic completed in 0.565 s wall time and 0.459 s CPU time, with 39.81 MiB sampled peak memory and 11.07 MiB of outputs. Twenty synthetic software checks and 2,665 saved-artifact checks support its implementation. These costs cover the entire small diagnostic and cannot be interpreted as robot deployment latency.

9 Discussion

The studies distinguish three questions that a single detection score cannot answer: whether unauthorized motion is detected, whether its physical effect persists, and whether the chosen clearance rule can recognize normal motion. The candidate's reduced benign alarm count initially appears favorable, but two plants have an unreachable clearing condition. A matched ordinary control removes this obstruction and is stronger on the recorded outcomes. The uncertainty-addition component consequently has no demonstrated benefit here.

Improved commissioning addresses one part of the problem. Longer ARX histories reduce normal prediction error and make Q2's condition reachable. The remaining failure on Q3, together with the perfect-reference calculation, shows that reference bias and measurement noise require separate treatment. A maximum of noisy residuals is not interchangeable with a bound on model error. Once it is added to another noisy residual, the available tolerance can be consumed before any genuine physical deviation is considered.

This suggests a practical design sequence: define the physical return target and trusted information; check the return condition under normal motion; then evaluate detection, persistence

and recovery jointly. Algebraic reachability, point feasibility and successful dwell probes provide different evidence. None alone establishes a safe reset policy. Statistical estimation, averaging and noise-aware observers are established tools for revising such a decision, and a replacement would need comparison against those tools rather than a new name for the present combination.

Study III makes this distinction quantitative. A windowed uncertainty control can certify normal residuals that an instantaneous control cannot, while still failing under correlated noise or faster within-tolerance variation. The limiting terms have different meanings: a noise term shrinks with effective sample size, whereas the current-state drift term increases with window length. Ignoring correlation improves some reported certificate fractions by weakening the assumption contract, not by demonstrating more reliable recovery. Useful deployment would require these uncertainty quantities to be justified from commissioning observations and plant knowledge; they were supplied by the generator in Study III.

The same-plant counterfactual also changes interpretation. In 21 intervention episodes, nominal-reference and actual-plant return times disagree. Measuring recovery against the detector's own nominal model can conflate model mismatch with remaining attack effects. Simulation permits the two trajectories to be separated exactly within the model. In a physical deployment, the counterfactual would itself be uncertain and would need independent justification.

9.1 Evidence scope

The evidence comes from three specified plant configurations in one simulator. Noise seeds repeat a physical condition; they are not independent robot acquisitions. Geometry, damping, lag and noise vary together, preventing causal attribution of a subgroup difference to a single factor. There are no contact-rich tasks, changeovers between manufacturing processes, field attacks or hardware measurements. The observed return tolerance has no established safety interpretation.

The attack comparison contains common-information controls and a restricted recent calibration component, rather than three full recent paper reproductions. The separate IMU study adds executable, common-input architectural comparisons, but its prefix is reused normal development, its compact models remain epoch-limited, and the exact original publication-run correspondence is unresolved. External-IMU baselines cannot be ranked fairly against protected-command monitors without resolving their input and protocol differences. This limits comparative-performance claims even where the internal ablations are informative.

The stages have different evidential roles. Study I prospectively evaluates a fixed candidate. Study II diagnoses its failure on new normal tasks. The post-study calculation explains a mechanism under explicit assumptions. Study III was specified before generating its residual traces, but it was motivated by the earlier failures and remains a simplified diagnostic. Its three noise realizations are paired across settings and are not independent robot acquisitions. All reported conditions are now exposed. A redesigned rule evaluated on them would be exploratory until tested under a new prospective protocol.

10 Conclusion

Alarm recovery depends on more than prediction accuracy. In the declared simulation, adding an empirical uncertainty allowance halves benign alarm observations relative to nominal fixed return but prevents timely clearance in two-thirds of detected-returned events. The matched control without that allowance performs better. Improved clean-data references reduce discrepancy without restoring feasibility across all plants. The Gaussian analysis identifies how calibration noise

consumes the return margin, while the covariance-aware window experiment shows a further trade-off between averaging and current-motion uncertainty. Its criterion is met on 20 of 36 constructed normal traces. The separate IMU comparison also retains a strong ordinary autoregressive control after longer CNN training. Together, these findings support evaluating uncertainty, information access and normal-return feasibility explicitly. They establish neither a universally effective reset policy nor superiority over CASPER.

A Command trajectories and reproducibility

For phase $\theta(s)$, amplitudes a_1, a_2 and offset ϕ,

$$q_1^c = a_1 \sin\theta + 0.025 \sin(2.3\theta), \tag{16}$$

$$q_2^c = a_2 \sin(\theta + \phi) + 0.018 \sin(1.7\theta). \tag{17}$$

Velocities are analytic derivatives multiplied by the causal time-warp rate. Study I tasks E, F and G use, respectively,

$$\begin{aligned} E: &\quad \theta = 0.63s, \quad (a_1, a_2, \phi) = (0.31, 0.35, 0.8), \\ F: &\quad \theta = 0.38s + 0.008s^2, \quad (0.26, 0.40, 1.2), \\ G: &\quad \theta = 0.83s + 0.18\sin(0.31s), \quad (0.36, 0.32, 0.4). \end{aligned}$$

Study II tasks H, I and J use the same respective phase forms, with base rates 0.52, 0.45 and 0.93, and amplitude/offset tuples $(0.28, 0.37, 1.4)$, $(0.34, 0.33, 0.7)$ and $(0.25, 0.38, 1.0)$. Authorized slowdown retains accumulated phase delay; unauthorized slowdown rejoins the approved schedule on restoration.

Commissioning episodes use phase rates 0.35, 0.7 and 1.1, with amplitudes $(0.22 + 0.04i, 0.34 - 0.025i)$ and offset $0.5 + 0.3i$, $i = 0, 1, 2$. They respectively use unchanged, authorized-offset and authorized-slowdown conditions. Normal calibration episode $i = 0, \ldots, 18$ uses rate $0.32 + 0.047i$, amplitudes $(0.23 + 0.012(i \bmod 7), 0.29 + 0.013(i \bmod 9))$ and offset $0.4 + 0.06(i \bmod 11)$, cycling through these benign conditions.

The retained reproducibility package contains configurations, source/input hashes, all fitted candidates, selection and calibration records, score seals, complete outcomes, dependency versions and resource logs. Study I is recorded as recovery v0.8; Study II as commissioning v0.9; Study III as return-bound v0.12; and the separate IMU study as CASPER convergence v0.11. They use Python 3.12.15 and NumPy 1.26.4 in isolated project environments; the CNN study additionally uses TensorFlow 2.16.2. Tables and figures are generated from saved result files, without fitting or simulation during manuscript production.

Study I completed in 229.20 s supervised wall time and 228.07 s worker CPU time, with 167.25 MiB sampled peak memory and 118.32 MiB of outputs. The worker's process-lifetime peak was 217.41 MiB. Study II used 14.75 s wall and 14.30 s CPU, with 113.97 MiB sampled peak memory, 117.66 MiB worker peak and 28.02 MiB of outputs. These are whole-run measurements, not per-observation deployment latency.

Fourteen synthetic checks preceded Study I; 23 preceded Study II. Post-run audits passed 9,610 and 10,293 checks, respectively, covering hashes, fitted artifacts, score alignment and independently recomputed metrics. The initial Study I launch failed before worker initialization because process monitoring was blocked. The initial Study II attempt stopped on a reporting-key collision after selection and before fresh generation. A separately recorded repair preserved byte-identical fits and selection before continuation. These attempts and the numerical replay environment are retained in the package. Software checks support reproducibility rather than physical validation.

B Separate common-input IMU prediction study

B.1 Release, chronology and permitted information

This supplementary study examines normal prediction under common inputs. It does not enter the command-aware attack ranking. The data are the received author-linked CASPER-1-and-2 provider-v2 nominal file `IMU_10Hz.csv`, from the public provider listing. Its SHA-256 is

`1221df9be80c229969f3f21f820837e68ad28f4ccb3949bf4013f00d6d69d62c`.

The file differs from the original CASPER-v3 archive; exact original-paper run identity is unresolved. The nine channels are accelerometer, gyroscope and magnetometer X/Y/Z. Timestamp units and acquisition-versus-arrival semantics remain unresolved, so the filename is not used to infer elapsed time or detection delay.

Training uses numeric rows 219–9184 and development uses rows 9185–12006, indexed from zero. The split precedes a trailing width-20 median filter. Each partition loses its first 19 filtered warmup observations. Training alone supplies the channel means and population standard deviations; constant or nonfinite channels are rejected. The clock must increase strictly, with no gap greater than five times the training median increment. All models use the same eligible targets after 755 history observations: 8,192 training and 2,048 development targets. The development target rows are 9959–12006 inclusive. A 755-residual aggregation leaves 1,294 aligned development endpoints. This prefix had already been used for development and is not an independent final test.

B.2 Architectures, fitting and model selection

The CASPER-style architecture follows the corrected manuscript's four valid one-dimensional convolutions, each with 32 filters, kernel size five and ReLU activation. Max pooling with width three follows each pair; flattening and a nine-output dense layer give 39,977 parameters for history length 755. The compact Edge-style architecture has two valid convolutions with eight filters and kernel size three, followed by width-three pooling and a nine-output dense layer: 1,369 parameters for history length 45. Both are independently written architectural implementations under a modified protocol. A zero-initialized output layer, batch size 64 and the present causal preprocessing are disclosed implementation choices, not proof of correspondence to a published checkpoint.

Training uses mean squared error, Adamax at learning rate 10^{-3}, chronological minibatches without shuffling, and seeds 17,42,2026. All models start afresh for this longer study; optimizer state was not transferred from the short pilot. The maximum is 100 epochs. A development improvement greater than 10^{-6} selects a checkpoint; five epochs without such improvement stop training. Per-fit CPU admission is limited to 240 seconds, with total ceilings of 1,200 seconds wall and 900 seconds CPU. No attack labels select training duration, model weights or score formulas.

Training-mean, persistence and AR(4) controls use the same target observations. The AR control fits standardized channels with ridge 10^{-3} on mean normal equations, leaving the intercept unpenalized. The reported MSE averages squared error over the nine standardized target channels and 2,048 development endpoints. Mean, persistence and AR run once; all stochastic seeds are reported separately.

B.3 Results and interpretation

All CASPER-style seeds stop by the declared patience rule after 27 epochs and select epoch 22, with MSE 0.017451–0.017703. This is 46.11–58.12% below their respective short-pilot errors. All

Table 6: Normal-development prediction on the common filtered IMU prefix. Epoch columns show completed and selected epochs. All nine outcomes are retained.

Predictor	Seed	MSE	Epochs	Stopping
Training mean	–	1.073987	–	Closed form
Persistence	–	0.019432	–	Closed form
AR(4)	–	0.006074	–	Closed form
CASPER-style CNN	17	0.017703	27/22	Patience
CASPER-style CNN	42	0.017451	27/22	Patience
CASPER-style CNN	2026	0.017458	27/22	Patience
Edge-style CNN	17	0.044730	100/100	Epoch limit
Edge-style CNN	42	0.043451	100/100	Epoch limit
Edge-style CNN	2026	0.042700	100/100	Epoch limit

compact seeds reach 100 epochs while improving, so their MSE 0.042700–0.044730 remains training-limited. AR(4) has the lowest error, 0.006074, on this prefix. Figure 4 retains the learning curves for every neural seed.

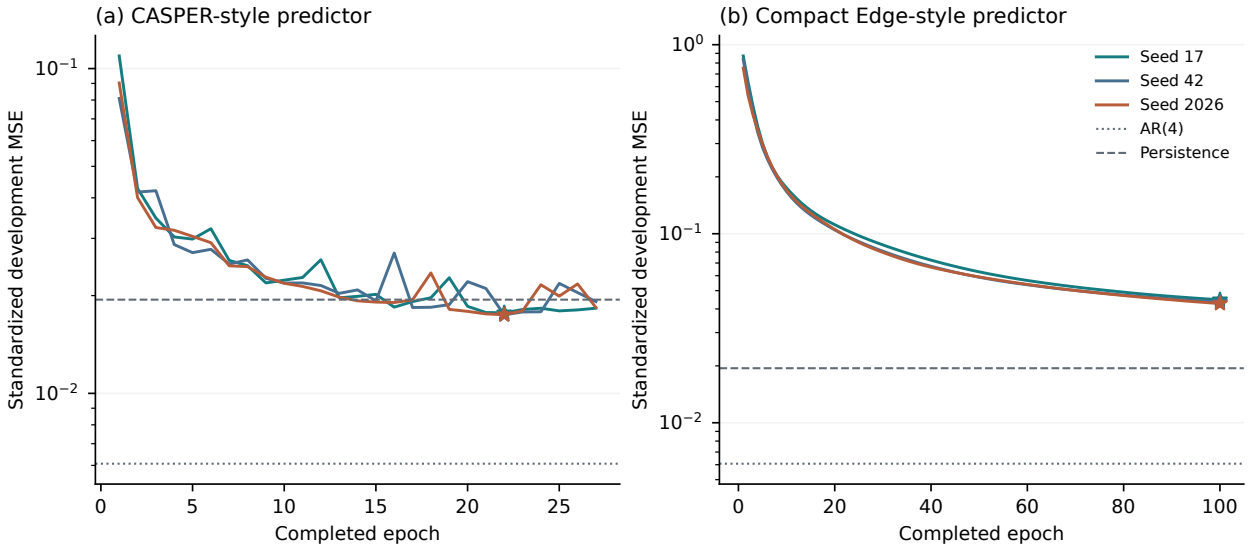


Figure 4: Normal-development learning curves from the saved training records. Local patience stopping supports greater training adequacy for the CASPER-style architecture; the compact models remain limited by the epoch ceiling. Horizontal control lines refer to the same filtered targets.

Median filtering induces dependence and changes the prediction task; the strong autoregressive result therefore cannot be transferred to unfiltered attack detection. Patience stopping is not proof of a global optimum, representative full-data convergence or original-paper reproduction. No attack threshold, F1, recovery score or test-label optimization was computed in this study. Improved normal MSE does not establish a more effective cyberattack detector.

The run used 764.794 s wall time, 753.252 s CPU, 553.05 MiB sampled peak memory and 2.68 MiB of outputs. Seventeen pre-run synthetic checks and 3,754 fresh-process artifact checks covered checkpoint reloads, score alignment and source identities. The hardware was the same macOS arm64 workstation; process threads and actual budgets are retained in provenance. No neural model was refitted while preparing this manuscript.

C Additional attack metrics

Table 7 uses unadjusted command-corruption labels. AP evaluates scalar-score ranking, not clearance quality. Unknown observations remain eligible, cannot count as detected attacks, and receive AP score −1, below valid nonnegative scores. Known coverage excludes unknown states. Alarm episodes per hour divide starts on wholly benign episodes by their eligible duration. Point rules can have frequent alarm starts despite a small alarmed-point fraction.

Table 7: Study I: additional metrics. All methods have 237,600 eligible observations and 135 intervention events.

Method	Precision	Recall	AP	Episodes/h	Missed	Known (%)
Persistence	0.203	0.033	0.130	480.8	8	100.0
ARX one-step	0.868	0.748	0.770	1426.3	27	90.0
Following point	0.878	0.747	0.770	793.9	27	90.0
Following hysteresis	0.841	0.735	0.770	99.0	27	90.0
Nominal hysteresis	0.706	0.719	0.587	84.8	27	90.0
Nominal fixed return	0.656	0.658	0.587	48.5	27	90.0
Learned point	0.658	0.748	0.589	1054.5	27	90.0
Learned hysteresis	0.628	0.747	0.589	62.6	27	90.0
Learned fixed return	0.641	0.747	0.589	115.2	27	90.0
Uncertainty return	0.318	0.747	0.589	26.3	27	90.0
Episode CP band	0.742	0.746	0.589	0.0	27	90.0
Matched no allowance	0.675	0.747	0.589	28.3	27	90.0
Matched no dwell	0.325	0.747	0.589	28.3	27	90.0

D Complete normal-reference comparison

Table 8 retains all distinct reference roles. Mean RMSE is computed over standardized state channels and averaged across episodes. Selected-reference and selected-ARX roles are identical and appear once. FIR models receive only past approved commands, while the one-step ARX uses observed history. Forced-latch clearance counts are normal diagnostics and must not be read as attack-recovery success.

Data and code availability

The attack/recovery episodes and residual traces are synthetic. The separate prediction study uses the author-linked CASPER-1-and-2 nominal release identified in Appendix B; original recordings remain subject to their provider's terms. Source code, configurations, hashes, fitted artifacts and complete outcomes are retained in the research reproducibility package. No public release of that package has been made. The manuscript source package contains the figure and table assets used in this article and does not redistribute raw recordings.

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Table 8: Study II: all distinct roles on the new normal tasks. Every row uses 29,700 eligible observations and 108 probes.

Plant	Reference	RMSE	Allowance	Feasible (%)	Clear/probes
Q1	ARX(4), free	0.00948	0.0553	99.80	108/108
Q1	Selected ARX, free	0.00934	0.0553	99.82	108/108
Q1	Selected FIR	0.01211	0.2322	0.00	0/108
Q1	ARX(4), one-step	0.00976	0.0649	97.92	108/108
Q1	Command as state	0.09584	2.1344	0.00	0/108
Q2	ARX(4), free	0.01020	0.1469	0.00	0/108
Q2	Selected ARX, free	0.00972	0.0716	93.32	108/108
Q2	Selected FIR	0.01693	0.3971	0.00	0/108
Q2	ARX(4), one-step	0.01036	0.1211	0.00	0/108
Q2	Command as state	0.10423	2.1528	0.00	0/108
Q3	ARX(4), free	0.01586	0.1913	0.00	0/108
Q3	Selected ARX, free	0.01439	0.1175	0.00	0/108
Q3	Selected FIR	0.02215	0.4596	0.00	0/108
Q3	ARX(4), one-step	0.01573	0.1807	0.00	0/108
Q3	Command as state	0.11251	2.2058	0.00	0/108

Competing interests

The author declares no competing interests.

Declaration of AI assistance

OpenAI Codex assisted literature checking, study design, programming, experiment execution, artifact verification and manuscript preparation. Numerical tables and figures were generated with Python and Matplotlib from saved experimental artifacts. No generative image model was used to produce the scientific figures.

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